

ORIGINAL RESEARCH

AI- and IoT-based hybrid model for air quality prediction in a smart city with network assistance

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Abstract

Air pollution is one of the biggest concerns in the world but it has not been paid much attention in developing countries. It is necessary to design models and methods to understand air pollution in developing countries to reduce the rate of pollution. This paper proposes an Internet of Things (IoT) and Artificial Intelligence (AI)-based hybrid model to predict the Air Quality Index (AQI) with a practical case study of the public data sets. The sensor node is deployed in the city to collect air quality data. Moreover, this sensor node connects to the cloud server for collecting data at the firebase real-time database through a WiFi/5G network embedded in the raspberry controller. Carbon monoxide (CO) and fine particular matter $PM_{2.5}$ sensors are integrated within a sensor node to monitor the AQI of the regions. A Kalman filter is also applied to remove unwanted noise from the data collected through the sensor node. Models namely Artificial Neural Network (ANN), Support Vector Machine (SVM), k-nearest neighbour (k-NN), Convolutional Neural Networks (CNN), Long Short Term Memory (LSTM), CNN-LSTM, ensemble model, and a proposed model, that is, CNN-LSTM-Bayesian optimization algorithm (BOA) model, have been utilised to predict the AQI. The performance evaluation of models is done through statistical parameters, such as mean absolute error (MAE), root mean square error (RMSE), coefficient of determination (R^2), and accuracy score on two different public data sets and compared with the baseline models. The performance of the CNN-LSTM-BOA model is better than baseline models in terms of above-mentioned statistical parameters as the accuracy reported is more than 97 %. This study can help predict the Air Quality Index and provide sufficient time to generate warning signals in the location.

KEYWORDS

5G Communication, AQI, Bayesian Optimization Algorithm, CNN, IoT, Kalman filter, LSTM

1 | INTRODUCTION

With the increase in industrialisation and vehicle emissions, the air around us is becoming polluted at a very fast rate [1]. Forest fires, agricultural burnings, volcanic eruptions, and many other activities lead to an increase in air pollution, a growing concern in major cities worldwide [2]. It is also responsible for respiratory problems like asthma and emphysema [3]. Although all worldwide countries have come up with policies and strategies to address the issue of air pollution, there is still a lack of

awareness, guidelines, and appropriate methodologies to reduce pollution [4]. Hence, there has been a continuous increase in the rate of pollution over the last few decades. In the past, several research works have been conducted highlighting the growth of air pollution with respect to its causes, factors leading to air pollution, its prevention and control. However, eradicating of air pollution from a local, regional, and global perspective is still a challenge [5–7]. One way to address the issue of air pollution is by considering the Air Quality Index (AQI), one of the most critical factors for evaluating, analysing,

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and reflecting the quality of the air [8, 9]. The AQI can determine the overall air quality using a scale ranging from 0 to 500. This scale is further categorised into six different levels of air quality, such as excellent, good, bad, light-moderate, moderate, and heavily polluted. Particular matter (PM) is a combination of solid particles and liquid droplets in the air. PM includes PM_{10} and $PM_{2.5}$. According to the American Health Association report, continuous exposure to a particular matter of size less than $2.5\ \mu\text{m}$ ($PM_{2.5}$) can lead to an increased risk of cardiovascular disease and decrease the lifespan cycle of normal human beings from several months to a few years [10]. These levels also indicate the impact of pollution on human health and suggest appropriate solutions for the safety of the people [11]. A lower AQI value indicates good air quality, whereas a higher number indicates poor. The Internet of Things (IoT) and artificial intelligence (AI) have made human lives easy by improved monitoring and improved control of operational processes in many domains [12–16]. Although IoT finds its use in many industries and applications, its ability to monitor pollution levels within cities by using specific devices and sensors makes it a necessary technology for pollution control [17, 18]. It works by placing the multiple stations in different city areas where the stations continuously upload and transmit data to the IoT cloud server [19]. At the time of data collection, sensors also generate noise in the data, which are essential to remove to increase the system's accuracy. A Kalman filter is one of the approaches used to reduce the noise from the collected sensor data in this paper. In 1960, a famous scientist Rudolf E. Kalman proposed research on linear filtering and problems based on the prediction [20]. The Kalman filter or linear quadratic estimation (LQE) is an essential algorithm for data fusion techniques. Kalman filter efficiency is directly dependent on features, such as the need for inexpensive computational equipment, proper framing of recursive properties, and matching required at the time of real-time implementation. Figure 1 describes the schematic overview of the Air Quality Prediction System. The cloud server collects the actual AQI values and stores them as a database to be used as a training data set in machine learning. The prediction of AQI can estimate the generalised environmental condition of a particular city so that effective measures can be devised to reduce pollution. There are some areas in which AQI monitoring is essential, which are: (1) the Industrial Indoor AQI system: It helps to maintain the suitable environmental conditions for the workers, (2) the Outdoor AQI system: Government and private entities can check real-time AQI of the city or particular area, and (3) PM Monitoring: These particles are small in size but can be easily inhaled and may affect on the human health, which can be mitigated with efficient monitoring.

Over the last few years, many researchers have proposed many solutions to forecast and control air pollution via IoT and AI techniques. In [21], the author proposed an IoT-based air quality monitoring system for urban and industrial areas. The design includes CO and NO_2 gas sensors and a server for real-time incident management. The low-rate monitoring system uses wireless communication, which also behaves as a

transceiver. The data values are sent to the mobile through the IoT server, and if there are any variations, the industrial management is updated accordingly. The work in [22] suggested a low-cost air pollution monitor for edge computing based on IoT. The basic idea is that sensors collect data, which are fed to edge computing devices for processing and analysis. The model performs well and reduces the computational burden by 70%. Data accuracy achieved is approximately 75%–80%, and the model is capable of reducing power consumption by 23%. In [23], researchers improvised a system deploying existing sensors for air pollution monitoring based on Narrow-band IoT (NB-IoT). The updated version of the design incorporates sensors for detecting carbon dioxide levels, ozone, particulate matter, nitrogen dioxide, and sulphur dioxide. Results indicate that the AQI measured is low, which is acceptable. The research in [24] suggested a low cost, open, and wireless IoT-based platform for monitoring air quality. Air quality is monitored by collecting data based on temperature, humidity, carbon monoxide, and nitrogen dioxide sensors. This collected data is analysed to comprehend the relationship between pollutant gases and the ambient parameters affecting air quality. Authors in [25] proposed the measurement of AQI using IoT. The study takes into account the process of sensing air from the local environment and analysing the concentration of $PM_{2.5}$ in the air as it is the most dangerous compound for human health. The results show a very high AQI for the location considered, which means that the air quality is not good. In [26], researchers presented a study on designing an IoT-based monitoring system for air quality. The proposed design is a mobile IoT-based platform, which analyses real-time data pertaining to carbon monoxide, smoke, and PM (particulate matter) levels. The system is user friendly and easy to handle. It can also be deployed easily. The work in [27] designed an IoT-enabled smart air quality monitor for indoor use. The system, which aims to monitor and improve air quality automatically, is built on a physical circuit and an interactive platform. Temperature and humidity sensors are also a part of this design. The interactive platform determines the AQI and sends alerts accordingly. The research, as mentioned earlier, is only based on data collection and data visualisation, but there is no AI-based prediction technique is applied. The research in [28] suggested a low power and reliable IoT-based platform for predicting $PM_{2.5}$. The study considers real-time monitoring of 10 air quality parameters and encourages remote monitoring and data storage. The wireless sensor node can easily integrate multiple sensors and interfacing pins [29]. Results show that the proposed platform is flexible, efficient, and accurate with a battery life of more than 31 months. Authors in [30] proposed an IoT-based system for monitoring air quality and particulate matter concentrations. The portable platform can check and measure air quality and PM for parameters, such as temperature, atmospheric pressure, dew point, and humidity. The digital dashboard provides real-time monitoring of data, which is uploaded to the cloud every 15 seconds. The device also collects data pertaining to $PM_{2.5}$, PM_{10} , AQI, and altitude. The data is analysed by comparing standardized parameters, such as $PM_{2.5}$ ($0\text{--}30\ \mu\text{g}/\text{m}^3$), PM_{10}

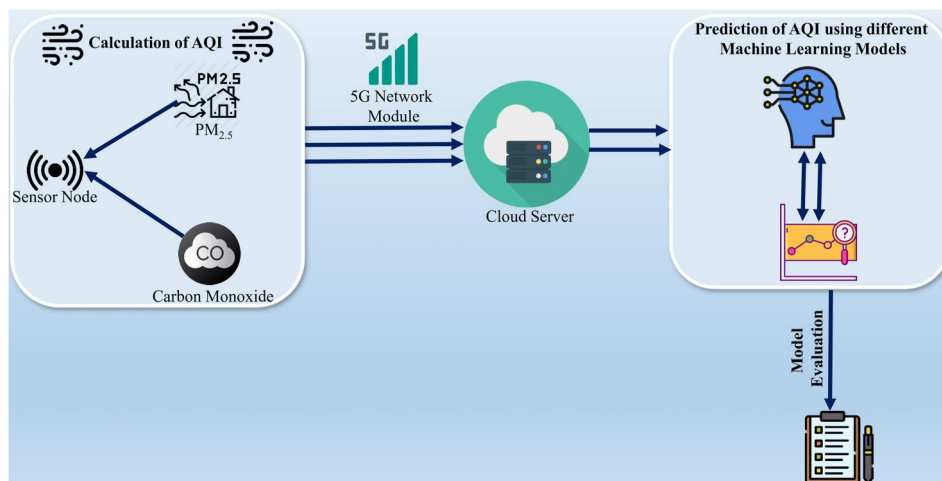


FIGURE 1 Schematic overview of the air quality prediction system

(0–50 $\mu\text{g}/\text{m}^3$), and Air Quality Index (0–50 parts per million (PPM)). Authors in [31] suggested a novel hybrid model for daily forecasting of AQI. The study takes into account the high-resolution data of various air pollution indices for AQI, $PM_{2.5}$, PM_{10} , carbon monoxide, etc. This is used to enhance the performance of 24,000 samples. Researchers have developed the real-time IoT solution to collect the air pollution parameters and evaluate their solutions based on the device level. Authors in [32–34] apply different models on public data sets to evaluate the performance of the model through different statistical parameters. In [35, 36], authors proposed lightweight and hybrid models to predict the AQI on different data sets. The AI models deployed in the above studies are in limited numbers to evaluate their experiment. In this study, a real-time IoT solution is integrated with AI techniques for the data collection and forecasting of the AQI. The integrated system helps monitor real-time AQI and forecast the AQI for the city.

This paper proposes a novel hybrid approach based on IoT and AI to measure a smart city's AQI index. For the study, we used a sensor node measuring carbon monoxide (CO) and $PM_{2.5}$ inside the city to collect air quality data and calculate the AQI of the regions, which are finally acquired at the cloud server. AI techniques, such as Artificial Neural Network (ANN), Support Vector Machine (SVM), k-nearest neighbour (k-NN), Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and CNN-LSTM, Ensemble, were used as baseline models. A novel technique proposed the CNN-LSTM-Bayesian optimization algorithm to predict the AQI. The models used in this study have been validated using statistical parameters, such as RMSE, MAE, R^2 , and model accuracy score. The main contributions of the paper are as follows:

- Design and development of the sensor node for the monitoring AQI data;
- Design a local smart gateway for data security;
- Application of Kalman filter to remove unwanted noise from the sensor node data;

- A novel model has been proposed for evaluation and forecasting, that is, hybrid CNN-LSTM-BOA. Seven existing techniques act as baseline models;
- Performance evaluation of models on the two public data sets;

The rest of the paper has been organised as follows: Section 2 discusses the background of the system. This section includes the related works followed by artificial intelligence techniques, and Section 3 illustrates the methodology and implementation. Section 4 discusses the results and analyses. This section also discusses performance evaluation. Section 5 concludes the proposed study and also discusses future work.

2 | BACKGROUND

2.1 | Artificial intelligence techniques

In this section, the baseline models used in this study are discussed. Table 1 represents the baseline models used in this study.

2.2 | Criteria for performance evaluation

In this study, performance evaluation of the models is performed using statistical methods, such as root mean square error (RMSE), mean absolute error (MAE), model accuracy score, and coefficient of determination (R^2). R^2 values vary between 0 and 1, which show the predictive capability of the deployed model. MAE and RMSE provide the error of the model used for training. Also, these evaluation technique values directly depend on model selection capability, such as minimum value means in which the model is as good and performs better. These statistical methods are calculated from Equations (1–3).

$$RMSE = \sqrt{\frac{1}{N} \sum_{K=1}^N (X_A - Y_A)^2} \quad (1)$$

TABLE 1 Baseline models used in the proposed work

Sr. no.	Model	Description
1	ANN	Artificial neural network (ANN) is an AI research domain using machine learning [37]. The researchers in [38] started research towards ANN and presented a mathematical model of the establishment and refinement of neural activities in the brain. Authors in [39] illustrated the learning procedure of the human brain with the use of reinforcement learning. The main purpose behind the research in the domain of ANN is to construct a machine learning-based system that can study and analyse the bio-electrical activities and biological model of the brain. ANN works as an intelligent recognition network machine [40]. In the proposed study, the first step is to upload the data and prepare for the ANN model. ANN regression is applied with an adaptive learning rate optimised for the model training. The models are evaluated through statistical parameters, such as R^2 , RMSE, MAE, and accuracy score.
2	SVM	Support vector machines (SVM) are the models based on supervised learning that is associated with learning algorithms that provide a valuable pathway to analyse data based on the classification and regression analysis [41]. The advantages of the SVM are (1) with a minimum number of features, the ability of SVM is more as compared to other model techniques; (2) model robustness is more; (3) the computational time of SVM is less as compared to the other techniques, such as the neural network; (3) SVM efficiency is more as compared to the other models [42–44]. In the SVM model, the initial steps are similar to the ANN model. The training of the SVM model with some specified parameters known as hyperparameters, such as $\gamma = 0.5$, kernel = 'rbf', true shrinking, and $\epsilon = 0.1$, helps to reduce the losses and increase the accuracy of the model.
3	k-NN	k-nearest neighbour's algorithm (k-NN) is a non-parametric method for data classification and regression [45]. Based on closest class neighbours, classification is done and k indicates the number of neighbours used to determine the class. For the last 50 years, k-NN techniques have been relied on by statisticians for machine learning research [46–48]. The algorithm is computationally efficient and easy to understand and implement. In the k-NN model, the initial steps are similar to the ANN model. The parameters configuration used in the k-NN model are five nearest neighbours, uniform weight, 'auto' algorithm for the ball tree or KD tree selection, and select 30 for leaf size. The performance evaluation is done through statistical parameters.
4	XGBoost	XGBoost is a gradient boosting decision tree (GBDT) optimised algorithm [49]. The main idea of this algorithm is that the performance of multiple decisions is better when compared to a single decision tree. It was developed by [50] to solve prediction problems and address issues related to overfitting or biasing of data. In the XGB model, the initial steps for data set processing are similar to the ANN model. At the time of the training model, some parameters are configured, such as $\gamma = 0$, learning rate = 0.09, maximum depth = 3, regression $\alpha = 0.75$, and regression $\lambda = 0.45$.
5	CNN	A convolutional neural network belongs to a class of deep neural networks and uses filters for performing convolution operations. The convolutional layer is the basic building block of a CNN [51]. The layer's parameters comprise a lot of learnable filters, which consolidate a less responsive field. During the forward pass, every filter is convolved over the width and height of the input volume, processing the dot product of the entries of the filter and the input and finally delivering a 2-dimensional activation map. Thus, the network learns through the filters and actuates when it identifies some particular kind of feature at some spatial position in the information. This is exactly like the reaction of a neuron in the visual cortex to a particular stimulus. In the CNN model, the initial steps are similar to other models. For the time series forecasting, a one-dimensional convolution layer is added with parameters like rectified linear unit, which is a linear function that helps to convert negative values to zero, maximum pooling layer that allows calculating the maximum value in the selected feature map, 50 densenet and adaptive learning rate optimiser.
6	LSTM	LSTM neural networks are based on the recurrent neural network (RNN) that can grasp the long-term dependencies through feedback connections [52]. The performance of the traditional RNN network is poor while solving feed-forward neural network problems due to a lack of memory, especially in the case of time series problems. To acquire short-term memory, RNN models use repetitive circles of connections in the hidden layers and collect informatics knowledge from the time series data. LSTM models were devised to overcome these issues, which help to store the informatics knowledge on the memory cells [53]. For the proposed study, the initial steps for the data set are similar. The model is trained by using some parameters, such as a sequential layer, dense layer 50, and stochastic gradient descent optimiser (SGD).
7	CNN-LSTM	Researchers proposed the CNN-LSTM model to improve the prediction of time series forecasting models [54–56]. In this hybrid model, the CNN model helps in the extraction of useful informatics knowledge and learn the internal structure of the time series data and the LSTM model is deployed to recognise both dependencies, such as long term and short term. CNN-LSTM model is categorised into two layers: 1) CNN and 2) LSTM. The CNN layer consists of the convolutions and pooling layers used to compute the complex mathematical equations to develop extracted features from the input data. The LSTM layer exploits the features extracted through CNN. In this study, the CNN model consists of two convolutions layers and one pooling layer, and the LSTM model consists of one neuron output layer.

TABLE 1 (Continued)

Sr. no.	Model	Description
8	Ensemble	Stacking presents the idea of meta learner, which is a substitute idea of voting, which is generally utilised by bagging. Bagging is a meta-algorithm dependent on ensemble machine learning uniquely intended for model robustness and performance improvement. The fundamental job of the stacking is to recognise or characterise the reliable model and help the meta learner to find an approach to incorporate the results of the best base learner. Level-1 models and level-0 models are the forecasts of the meta-model information and base model individually. The stacked learning is utilised for the classification, and an event is presented to the level-0 model so that each event predicts a class value. These predictions are additionally given to level-1 and are then consolidated to build the final prediction. Training the stacking model is precarious although not a difficult task. The training steps of the stacking model are like the k-fold validation [57]. Figure 2 addresses the ensemble machine learning procedure for the proposed framework. SVM, ANN, and k-nearest neighbours may be considered as base learners, and XGBoost may be considered as a meta learners. In the base learner, three unique models are utilised, for example, SVM, ANN, and k-NN, and are trained based on the training data set. Each model gives predictions, and these predictions structure another training set for the XGBoost model and the model may be considered a meta-learner. Finally, XGBoost trains another data set from the base learning and predicts the values that define accuracy score, RMSE, MAE, and R^2 .

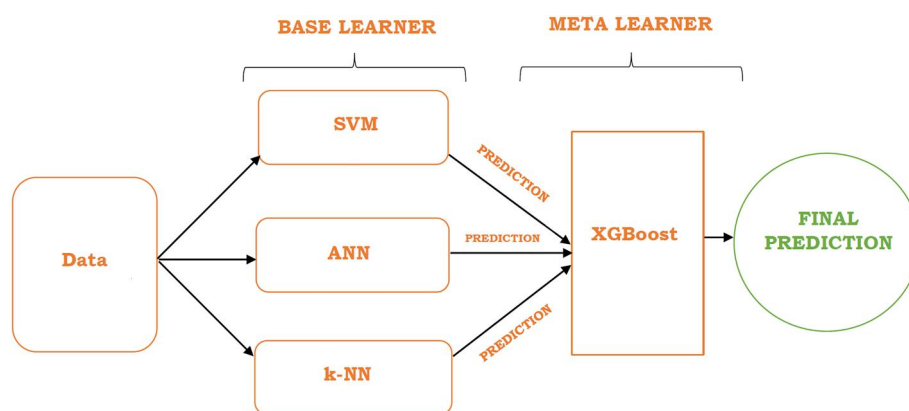


FIGURE 2 Ensemble machine learning model

$$MAE = \frac{1}{N} \sum_{K=1}^N |X_A - Y_A| \quad (2)$$

$$R^2 = 1 - \frac{\sum_{K=1}^N (X_A - Y_A)^2}{\sum_{K=1}^N (X_A - \frac{1}{N} \sum_{K=1}^N X_A)^2} \quad (3)$$

where X and Y are actual and predicted values, respectively.

3 | METHODOLOGY

3.1 | Proposed system architecture

The architecture for the proposed solution is categorised into five layers described in Figure 3. In layer 1, sensor nodes equipped with pollution detector sensors, such as $PM_{2.5}$ and CO, a controller, and a 5G Network/Wi-Fi module, are deployed inside the city to measure air pollution. Also, a Kalman filter is applied to remove the noise from the data acquired by the sensor. In layer 2, sensor nodes are connected to the Google cloud server via a smart gateway, and this layer is also used to set up connectivity between sensor nodes and the

cloud server. In layer 3, the collected data is exported from the firebase in the form of a JavaScript Object Notation (JSON) file. Before applying these techniques, missing values or inappropriate data are removed from the data set. In layer 4, machine learning models are applied to the existing data sets. These machine learning models include the currently existing models, such as ANN, k-NN, SVM, LSTM, CNN, CNN-LSTM and Ensemble (discussed in section 2), and the proposed model, such as the CNN-LSTM-BOA [58]. Layer 5 is to measure the performance of these models through statistical methods, such as RMSE, MAE, and R^2 . Moreover, CNN-LSTM-BOA performance compared with the existing models (discussed in section 4).

3.2 | Sensor node deployment

Sensor node is a portable device equipped with $PM_{2.5}$ and CO sensors to monitor the AQI values and upload the data on the cloud server. Figure 4 represents the schematic of the proposed sensor node. A portable sensor node is used to collect the data from different locations inside Da Nang City, Vietnam. In this study, a sensor node is used to monitor $PM_{2.5}$ and

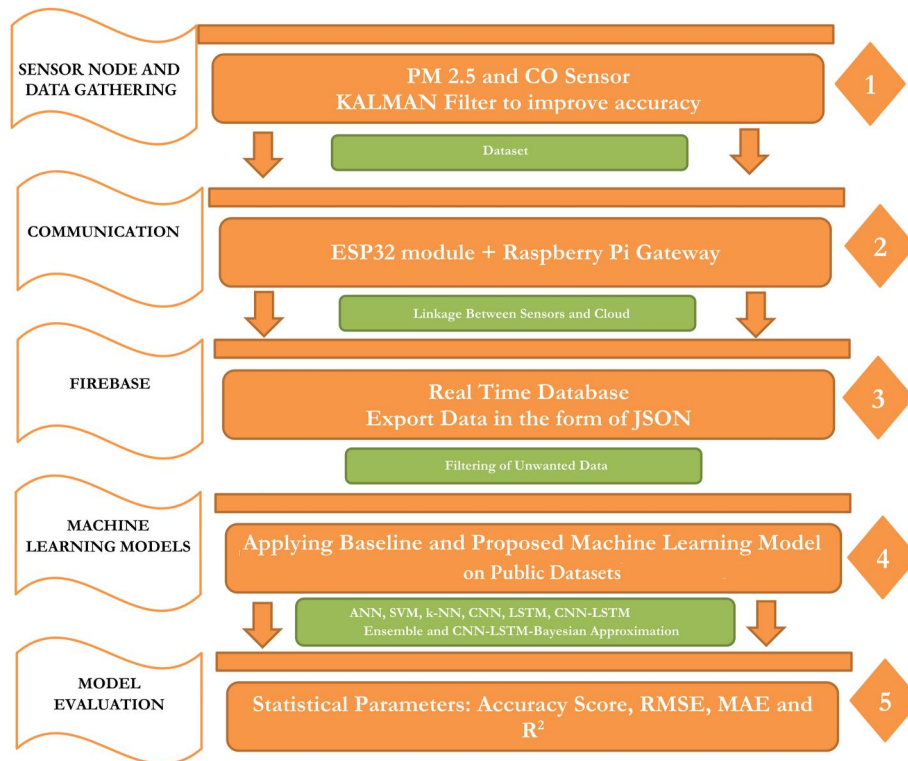


FIGURE 3 Architecture for the proposed solution

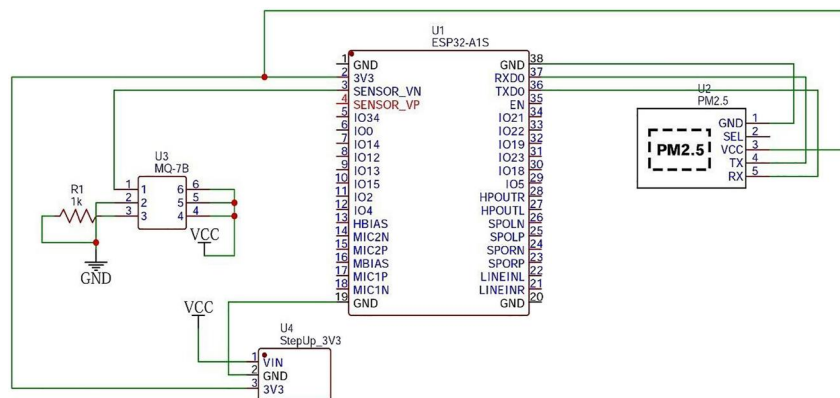


FIGURE 4 Schematic of sensor node

CO concentrations. These concentrations can be converted to AQI values. AQI is a numerical scale that describes the air quality and its health effects on human life [59, 60]. Higher values of AQI correspond to degraded air quality and more severe human health problems. The value of the AQI is directly proportional to the severity of human health problems. Some gases like CO, SO₂, and O₃ pollutants are the primary gases measured in terms of the AQI. Moreover, PM and O₃ play a significant role in increasing the risk of environmental pollution and elevating health problems. The workflow of the Kalman filter is processed into two steps, the first one is Prediction and the second one is Update and Correction. The mean and variance of the result are cyclically overridden by the filter. As long as the data do not differ too far from the expected value, the filter will always be confident in its location.

Even though the data are noisy, the filter gets better over time to assure accuracy (normal distributions go narrower and higher) since the measured values (in update) suit the anticipated ones (by predict) rather well. The Kalman filter is used in the suggested system along with microcontroller processing. The Kalman filter is used in the suggested system along with microcontroller processing.

3.2.1 | Data processing

In the monitoring and processing part of the sensor node, PM_{2.5} and CO are deployed to monitor the concentration value of the pollutant and further process this value to calculate the AQI. In this sensor node, Plantower and MQ7 detectors are

used to monitor $PM_{2.5}$ and CO, respectively. $PM_{2.5}$ is an atmospheric particulate matter with a size of less than 2.5 μm , which is nearly 3% of human hair size [61]. If the particle size is less than 10 μm , it is also referred to as PM_{10} . Also, if the particle size is less than 10 μm , it is considered as a fine particle. PMS3003 is a digital sensor that can be used to capture suspended particles in the air and transform them into digital outputs for further processing with the aid of a controller or processor [62]. The main principle behind this sensor is laser scattering. This sensor captures scattering light at a particular angle and monitors the changes in the curve of scattering light with light. The operating voltage of the sensor is 5 V, and the data transfer logic level is 3.3 V. Carbon monoxide (CO) is a toxic, colourless, odourless gas produced from improper combustion of petrol, diesel, natural gas, coal, and wood. CO molecular weight is nearly equal to the air that provides a highly free nature to mix with air and move along with it. CO gas is non-detectable to humans from any senses, such as taste, visibility, and smell. CO is directly soluble in the water and leads to high health risks, such as heart disease or respiratory problems that can move living creatures towards dying situations. MQ7 is an analogue-based gas sensor that is powerful enough to detect carbon monoxide gas in the air and transform the output into an analogue voltage [63]. The concentration of carbon monoxide gas generally lies from 10 ppm to 10,000 ppm. The operating temperature of MQ7 sensor varies from -100 to 500 C, and the operating voltage and current are 5 V and less than 150 mA, respectively. ESP32 is a low cost, low power microcontroller chip integrated with Wi-Fi, and Bluetooth [64]. The Control Processing Unit of ESP32 is a dual-core 32-bit microprocessor operated on 160 MHz or 240 MHz and is equipped with a memory of 520 kB Static Random-Access Memory (SRAM). Security features like Work-From-Anywhere (WFA), Wi-Fi Protected Access (WPA/WPA2), and WLAN Authentication and Privacy Infrastructure (WAPI) are supported in ESP32. Moreover, it also enables cryptographic hardware acceleration. In the proposed study, ESP32 fetched data from the PMS3003 and MQ7 and processed it. The AQI is calculated from the fetched data. Sensor calibration is an essential part of any implementation. Sensor calibration is a set of instructions to make instruments or sensors more accurate and error-free. Two sensor calibrations, such as MQ7 and $PM_{2.5}$, are done with different gas concentrations and particles present in the air as shown in Table 2.

For the true AQI, there is a need to calculate the values for the CO and $PM_{2.5}$ separately through the mathematical equations [65] as shown in Equations 4 and 5:

For CO,

$$AQI_{CO} = \frac{I_{v+1} - I_v}{BP_{v+1} - BP_v} (C_{co} - BP_v) + I_v \quad (4)$$

For $PM_{2.5}$,

$$AQI_{PM_{2.5}} = \frac{I_{v+1} - I_v}{BP_{v+1} - BP_v} (Nowcast_{PM_{2.5}} - BP_v) + I_v \quad (5)$$

$$AQI = \text{Max}(AQI_{CO}, AQI_{PM_{2.5}}).$$

Whereas C_{co} and $Nowcast_{PM_{2.5}}$ are concentrations of CO and $PM_{2.5}$, respectively. BP_{v+1} , BP_v , I_{v+1} , and I_v are the concentration breakpoints $\geq C_{co}$, $PM_{2.5}$, $\leq C_{co}$, and $PM_{2.5}$, AQI values correspond to BP_{v+1} ($PM_{2.5}$, CO), AQI value corresponds to BP_v ($PM_{2.5}$, CO), respectively. True AQI is the maximum value of AQI_{CO} or $AQI_{PM_{2.5}}$. For performance evaluation, initially, the device collected the sensor concentration data for the 10 days and calculated the AQI through the sensor node as well as manually by substituting concentration values of $PM_{2.5}$ and CO in the Equations 4 and 5. The next step is verifying the sensitivity of sensors in different environments, such as automobile exhaust, vehicle crowd, beach area, and industrial environment. In the final step, sensor concentration values are matched with other bench-marked sources, such as the IQ Air Index and [Air-quality.com](https://airquality.com) ($PM_{2.5}$ and CO concentration). Finally, the sensor node is ready to collect the AQI data of the concerned city.

3.3 | Smart gateway

The proposed sensor node is deployed at a particular location inside the city that is connected through the Smart Gateway as shown in Figure 5. Smart Gateway is implemented in the raspberry for setting up a secure and reliable connection for the air monitoring device with the cloud server through WiFi or SIMCom 5G module, compatible with raspberry [66, 67]. The main purpose behind the IoT gateway is to enable or set up communication between the nodes or between the node and the cloud. It also secures the communication link between the device and cloud server and maintains reliable security [68–71]. Generally, the gateway can be defined as a hardware device that facilitates the source data to the destination [72]. Visualisation,

TABLE 2 MQ-7 and $PM_{2.5}$ sensors calibration

S.no.	MQ-7 calibration		$PM_{2.5}$ Calibration(PMS3003)	
	CO concentration %	Voltage generated (v)	Particles %	Concentration($\mu\text{g}/\text{m}^3$)
1	0–20	0.24	0–20	5
2	20–50	3.02	20–50	14
3	50–70	3.98	50–70	40
4	70–100	4.60	70–100	60

short-term data retention, managing network security, and system diagnosis are some essential features of the Smart Gateway. Firebase is based on Google web development and mobile platform [73]. In the Smart Gateway, two servers host the data: 1) apache server (webserver) and 2) node.js server (IoT gateway). The workflow behind the IoT services is as follows:

- Deployed sensor nodes detect the variations in the values of the sensors and consider them as events and transmit them to the smart gateway.
- Gateway collects the data from the sensors and pre-filters the received values, and refines them. Every sensor value is compared with the existing threshold values and stamped with date and time. Then, the registered events are moved to the Google cloud services.
- Google cloud processes these events and stores them in the cloud services.

A web server provides the web pages that help generate the credentials and transfer them to the web browser. These credentials are moved towards the IoT gateway server through web servers. Apache server works as a bridge called reverse proxy between the IoT gateway and the browser. The credentials are moved from the web browser to the apache server and further to the node.js server with the help of HTTP (Hypertext Transfer Protocol) post request. The HTTP server is also deployed in the IoT gateways that lead to the

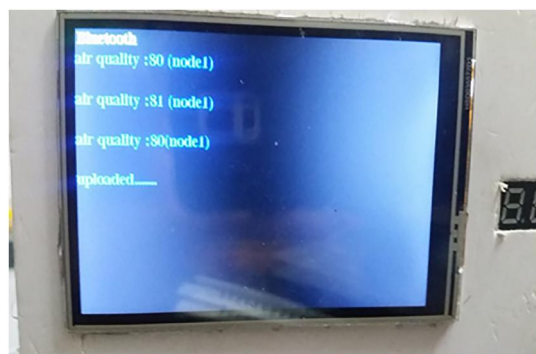


FIGURE 5 AQI values on the Sensor Node

authentication of the firebase via credentials. The real-time database relies on the cloud-hosted database. This database is also used to store data as JSON. It is connected with the client for real-time synchronization. The main feature of a real-time database is that the same database can be shared across multi-platform devices like iOS and Android. Further, it works online as well as in offline modes. For the offline mode, data may be stored on a disk. When it is connected (online mode), it may synchronize with the respective state of the firebase server. JSON is an open standard format file, which consists of a language-independent data format [74]. It uses human-friendly texts for sharing data objects that incorporate data-type arrays and value pairs. The firebase database, which stores JSON objects, is usually created as a JSON tree. Every updated data in the JSON tree generates a key-enabled node that consists of the USER ID. The data collected can be exported in the form of JSON files. This data can be used for the application of machine learning models to create better predictions on the data in a city or smart city [13].

3.4 | Materials and processing

For the material and processing, two different data sets are considered: (1) Data set 1: Public Data set based $PM_{2.5}$ Concentration (2) Data set 2: Public Data set based on AQI.

3.4.1 | Data set 1

Data set 1 is retrieved from the UCI machine learning repository based on Beijing $PM_{2.5}$ concentrations [75, 76]. The collection range of the data set is from 2010 to 2014, as shown in Figure 6, and contains 43,824 samples. The main parameters of the data set include years, month, date, hour, $PM_{2.5}$ concentration, dew points, temperature, pressure, combined wind direction, cumulated wind speed, cumulated hours of snow, and cumulated hours of rain. The duration of every registered sample is 1 h. This data set contains redundant values, which is identified and removed during the preprocessing techniques. The reason behind using the $PM_{2.5}$ data set is to evaluate the performance of the proposed models.

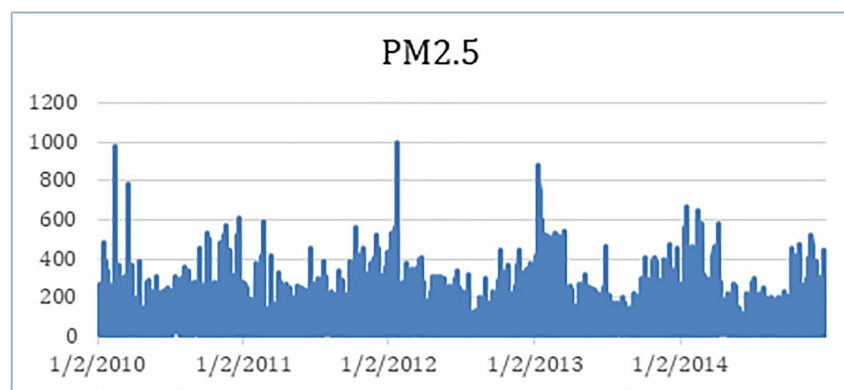


FIGURE 6 $PM_{2.5}$ concentration value from 2010 to 2014

3.4.2 | Data set 2

Data set 2 is based on values derived from AQI Index India (Chennai city) and has been retrieved from Kaggle [77]. This data set contains 335,023 samples, which include the date, time, $PM_{2.5}$, NO, NO_2 , NO_x , NH_3 , CO, SO_2 , O_3 , Toluene, and AQI. The data collection range lies between 2015 and 2019 as shown in Figure 7. The duration of every registered sample is 1 h, and it also contains blank values. The preprocessing techniques applied to eradicate the blank values are similar to other data sets.

3.5 | Proposed artificial intelligence techniques

3.5.1 | CNN-LSTM-BOA

The Bayesian optimization algorithm (BOA) is a highly efficient framework for optimising black-box function globally without utilising the gradient information and grasping the function distribution through formalising the optimization problem [58, 78]. BOA utilises the probabilistic model to capture the unrecognised functions, and for that, the Gaussian process is always the best choice for the model. The main principle behind the BOA is to figure out the Gaussian process model through its mean and covariance function $f(U)$. Suppose $U \in X^d$ is the initially optimised vector of the decision variable from randomly selected sample points of the decision variable, and U_i and V_i (corresponding cost) are enumerated to combine the pair set of input-output. This set is employed to figure out a Gaussian process model and probabilistic model for this set in Equation 6:

$$O(V|U_i, V_i) = Q\left(V|h(U_i)^T \cdot \beta + \mu(U) + k(U, \theta), \sigma^2\right) \quad (6)$$

where O is the output of the model, the conditional probability of observing V in given U_i and V_i , Q is the posterior probability of V , h is the base function, β is its coefficient vector, μ , k and σ^2 are the means, covariance, and noise standard deviation. In machine learning algorithms, hyperparameters are the pivotal

part. Using traditional methods, such as grid or random search, the tuning of hyperparameters can be a complex process. Nowadays, BOA is popular among machine learning models due to its efficiency in tuning the hyperparameters. It can also be used as an autonomous algorithm to handle and tune hyperparameters. The hyperparameters tuned in the CNN-LSTM-BOA are activation, neural network layers, dropout layers, and learning rate. These parameters have minimum and maximum values to tune the hyperparameters. Similarly, in the case of activation, different activators are considered 'relu', 'tanh', 'linear', 'selu', and 'elu'. Figure 8 represents the workflow of the proposed CNN-LSTM-BOA. In the first step, the CNN-LSTM model and hyperparameters are initialised. After the successful training of the model, a trained CNN-LSTM model is ready. In the next step, model prediction is collated with the validation RMSE. The main aim of BOA is to reduce the RMSE validation until an optimal set of hyperparameters is not achieved. After achieving the optimal hyperparameter set, the CNN-LSTM model is tuned with these sets and compared with the test RMSE values. The main advantage of using the BOA with CNN-LSTM is selecting the appropriate parameters for better prediction. However, the model latency is higher on a more extensive data set than CNN-LSTM.

The first step for the proposed algorithm is to load the data set and apply the label encoder on the data set that helps

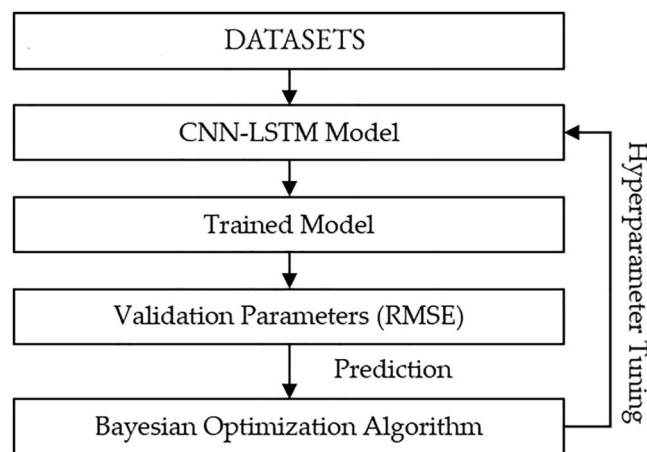


FIGURE 8 Workflow of the CNN-LSTM-BOA model

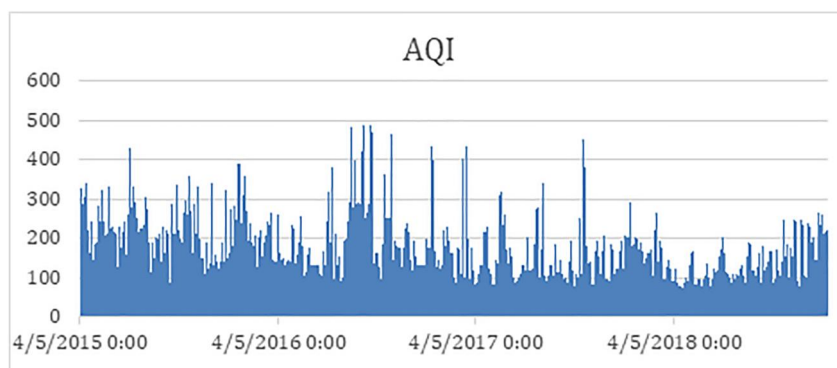


FIGURE 7 AQI value from 2015 to 2020

convert the labels into numeric values, followed by normalisation to scale the data features between 0 and 1. The next step includes reframing to convert data from the time series to supervised learning. Up to this point, the processing steps for every algorithm used in this study are the same. In the CNN-LSTM-BOA algorithm, multi-variable input data is split into the training and testing sets. After successfully preparing the data set, the CNN-LSTM model is designed, and the performance is evaluated using statistical parameters. In the last step, the BOA algorithm is applied for the hyperparameter tuning. The primary purpose of BOA is to tune and search hyperparameters that provide the better results.

4 | RESULT AND DISCUSSION

For evaluating the performance of the proposed work, machine learning models are used, such as ANN, SVM, k-NN, CNN, LSTM, CNN-LSTM, ensemble technique, and hybrid CNN-LSTM-BOA model. The evaluation for these models is done using statistical parameters, such as R^2 , RMSE, MAE, and accuracy scores and validated through Python library sklearn metrics [79]. Data set 1 and Data set 2 contain 43,824 and 335,023 samples, respectively. These data sets are divided into training and testing subcategories in the ratio of 80:20%. The epochs considered in the experiment is 50.

The result featured the predicted value of AQI obtained from the training, testing, and validation of ANN, SVM, k-NN, CNN, LSTM, CNN-LSTM, ensemble, and hybrid CNN-LSTM-BOA that have mutuality with the experimental value of AQI. Table 3 and Table 4 represent the performance evaluation of different models on Data set 1 and Data set 2 respectively.

In this study, the purpose of evaluating the two data sets is to verify the performance of the proposed model for the prediction of AQI of a city or smart city. Sometimes, the accuracy of the model is very high at the time of training, but the overall model performance can be low or poor. In this condition, models face an over-fitting problem. Therefore, there is a mandatory analysis of the other parameters, such as R^2 , MAE, and RMSE, for measuring the actual accuracy of the model. LSTM model is one of the best models, especially in time series

forecasting. Although LSTM's accuracy is comparable to other baseline models, such as ANN, k-NN, SVM, and CNN, it is less than the CNN-LSTM model. To overcome these concerns, the ensemble model is used with XGBoost meta learner. The performance of the ensemble model is entirely satisfactory as compared to the CNN and other baseline models except for LSTM, CNN-LSTM, and CNN-LSTM-BOA. In CNN-LSTM, the overall performance of this model seems to be better than other base models, but there is also the existence of the performance variation. To ensure stable performance, the CNN-LSTM-BOA model is proposed to overcome the problem of variation. The proposed model performs better for both data sets than the other baseline models and the ensemble model. Specifically, the performance evaluation of the proposed model is up to 50 epochs, and the variation of the RMSE values is not high.

As per Tables 2 and 3, the accuracy score performance of the CNN-LSTM-BOA model on data sets 1 and 2 are 0.981 and 0.973, respectively. The performance evaluation of ensemble model in terms of accuracy score: R^2 : MAE: RMSE for data set 1 and 2 are 0.977:0.968:15.4625:24.1207 and 0.967:0.949:23.6978:37.2059, respectively. The CNN-LSTM-BOA model performed well on data sets 1 and 2 compared to the ANN, k-NN, SVM, LSTM, CNN, CNN-LSTM, and an ensemble model. Data sets 1 and 2 are based in Beijing (China) and Chennai (India), respectively. In the case of data set 2, ensemble performance is better as compared to the CNN-LSTM and LSTM models. In data set 2, the performance for ensemble models is better as compared to LSTM. The performance variation exists in the different data sets, but in the case of the proposed model CNN-LSTM-BOA, it performs well, and there is no variation in data sets 1 and 2. Figures 9 and 10 represent the performance evaluation of CNN-LSTM and CNN-LSTM-BOA models on public data sets.

5 | CONCLUSION

This paper proposed an IoT-based real-time monitoring system for air quality. A Kalman filter is applied to remove noise from the data collected by the sensor node at the time of data collection. Proposed models, such as CNN-LSTM-BOA, and

TABLE 3 Performance evaluation of different models on Data set 1

Model	Accuracy score	R^2	MAE	RMSE
ANN	0.818	0.886	22.4391	39.6554
SVM	0.869	0.822	38.7967	51.0213
k-NN	0.936	0.928	17.6982	32.5691
CNN	0.957	0.942	16.6502	24.4872
LSTM	0.969	0.931	15.8970	23.9681
CNN-LSTM	0.972	0.959	14.8967	23.1369
Ensemble	0.977	0.968	15.4625	24.1207
CNN-LSTM-BOA	0.981	0.964	12.0923	21.6932

TABLE 4 Performance evaluation of different models on Data set 2

Model	Accuracy score	R^2	MAE	RMSE
ANN	0.897	0.819	40.9067	46.5489
SVM	0.928	0.889	24.8768	34.1489
k-NN	0.881	0.824	34.6985	41.2346
CNN	0.913	0.898	31.8150	29.8596
LSTM	0.955	0.915	19.8975	31.5289
CNN-LSTM	0.978	0.936	26.6985	30.8562
Ensemble	0.967	0.949	23.6978	37.2059
CNN-LSTM-BOA	0.973	0.951	17.9656	27.9843

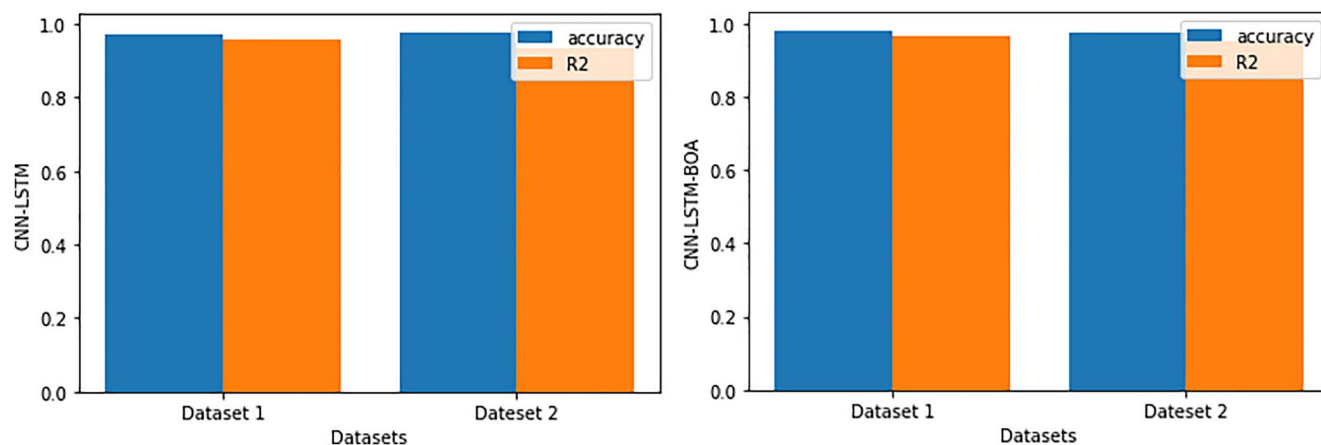


FIGURE 9 Accuracy and R^2 on different data sets

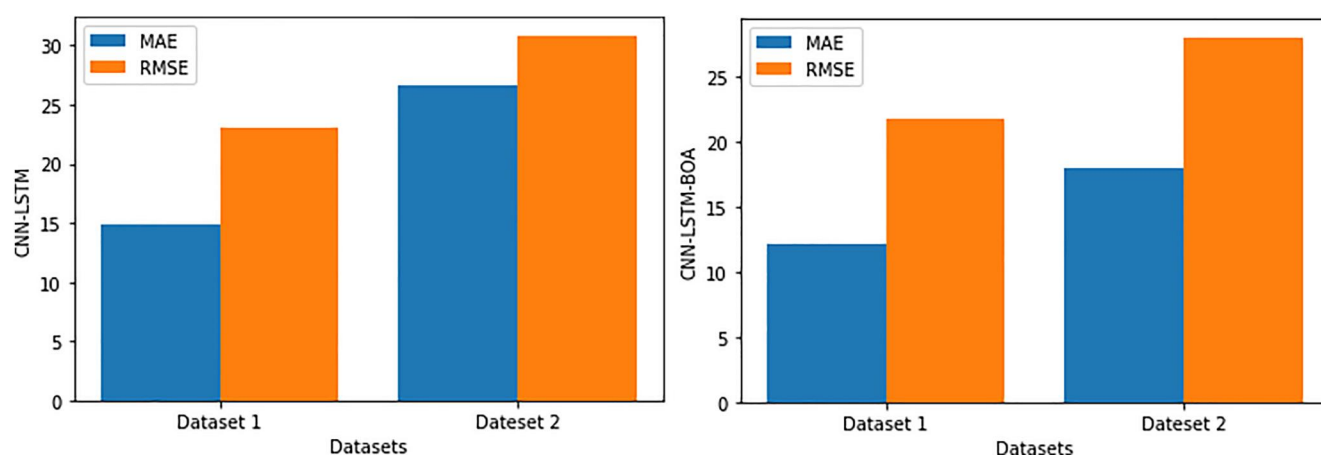


FIGURE 10 MAE and RMSE on different data sets

other baseline models, such as ANN, SVM, k-NN, CNN, LSTM, CNN-LSTM, and Ensemble model, are tested on the public data sets in this paper. Results showed that the baseline LSTM model is good for the prediction, but there is a variation in outcomes for both data sets. The validation of results is done through statistical parameters, such as RMSE, MAE, R^2 , and accuracy score. The results suggest that the proposed model performs better than other baseline models. Future work under the above-mentioned architecture is planned to collect the extensive data set from the sensor nodes and apply the proposed AI techniques to evaluate the performance through the statistical parameters and compare the resulting outcome with the public data sets.

CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in 1. UCI Machine Learning repository at [<https://archive.ics.uci.edu/ml/datasets/Beijing+PM2.5+Data>]. 2. Kaggle repository at [<https://www.kaggle.com/rohanrao/air-quality-data-in-india>].

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REFERENCES

- Wang, S., et al.: Strategizing the relation between urbanization and air pollution: empirical evidence from global countries. *J. Clean. Prod.* 243, 118615 (2020). <https://doi.org/10.1016/j.jclepro.2019.118615>
- Michaud, J.P., et al.: Volcano related atmospheric toxicants in Hilo and Hawaii Volcanoes National Park: implications for human health. *Neurotoxicology.* 26(4), 555–563 (2005). <https://doi.org/10.1016/j.neuro.2004.12.004>
- Ma, Q., et al.: Understanding the knowledge gaps between air pollution controls and health impacts including pathogen epidemic. *Environ. Res.* 189, 109949 (2020). <https://doi.org/10.1016/j.envres.2020.109949>
- Shanmugam, G.S., et al.: Smart green resource conservation approach for smart IoT cloud. *J. Comput. Theor. Nanosci.* 15(6-7), 2069–2075 (2018). <https://doi.org/10.1166/jctn.2018.7409>
- Zhu, L., et al.: Do economic activities cause air pollution? Evidence from China's major cities. *Sustain. Cities Soc.* 49, 101593 (2019). <https://doi.org/10.1016/j.scs.2019.101593>
- Sharma, R., et al.: Inferring air pollution from air quality index by different geographical areas: case study in India. *Air Quality, Atmosphere & Health.* 12(11), 1347–1357 (2019). <https://doi.org/10.1007/s11869-019-00749-x>
- Zhang, H., et al.: Air pollution and control action in Beijing. *J. Clean. Prod.* 112, 1519–1527 (2016). <https://doi.org/10.1016/j.jclepro.2015.04.092>

8. Kyrkilis, G., Chaloulakou, A., Kassomenos, P.A.: Development of an aggregate Air Quality Index for an urban Mediterranean agglomeration: relation to potential health effects. *Environ. Int.* 33(5), 670–676 (2007). <https://doi.org/10.1016/j.envint.2007.01.010>
9. Zhao, B., et al.: Evolution and comparative assessment of ambient air quality standards in China. *J. Integr. Environ. Sci.* 13(2–4), 85–102 (2016). <https://doi.org/10.1080/1943815x.2016.1150301>
10. Güçlü, Y.S., et al.: Air quality (AQ) identification by innovative trend diagram and AQ index combinations in Istanbul megacity. *Atmos. Pollut. Res.* 10(1), 88–96 (2019). <https://doi.org/10.1016/j.apr.2018.06.011>
11. Zhu, G., et al.: Health burdens of ambient PM_{2.5} pollution across Chinese cities during 2006–2015. *J. Environ. Manag.* 243, 250–256 (2019). <https://doi.org/10.1016/j.jenvman.2019.04.119>
12. Rani, S., et al.: Security and privacy challenges in the deployment of cyber-physical systems in smart city applications: state-of-art work. *Mater. Today Proc.* 62, 4671–4676 (2022). <https://doi.org/10.1016/j.matpr.2022.03.123>
13. Rani, S., et al.: Amalgamation of advanced technologies for sustainable development of smart city environment: a review. *IEEE Access.* 9, 150060–150087 (2021). <https://doi.org/10.1109/access.2021.3125527>
14. Kataria, A., Ghosh, S., Karar, V.: Data Prediction of Optical Head Tracking Using Self Healing Neural Model for Head Mounted Display (2018)
15. Puri, V., Kataria, A., Sharma, V.: Artificial intelligence-powered decentralized framework for internet of things in healthcare 4.0. *Transactions on Emerging Telecommunications Technologies*, e4245 (2021). <https://doi.org/10.1002/ett.4245>
16. Patel, D., et al.: Network anomaly detection inside consumer networks—a hybrid approach. *Electronics.* 9(6), 923 (2020). <https://doi.org/10.3390/electronics9060923>
17. You, I., et al.: An enhanced LoRaWAN security protocol for privacy preservation in IoT with a case study on a smart factory-enabled parking system. *Sensors.* 18(6), 1888 (2018). <https://doi.org/10.3390/s18061888>
18. Sharma, V., Kumar, R., Kaur, R.: UAV-assisted content-based sensor search in IoTs. *Electron. Lett.* 53(11), 724–726 (2017). <https://doi.org/10.1049/el.2016.3487>
19. Dhingra, S., et al.: Internet of Things mobile–air pollution monitoring system (IoT–Mobair). *IEEE Internet Things J.* 6(3), 5577–5584 (2019). <https://doi.org/10.1109/iot.2019.2903821>
20. Kalman, R.E.: A New Approach to Linear Filtering and Prediction Problems (1960)
21. Saikumar, C.V., Reji, M., Kishoreraja, P.: IOT based air quality monitoring system. *Int. J. Pure Appl. Math.* 117(9), 53–57 (2017)
22. Idrees, Z., Zou, Z., Zheng, L.: Edge computing based IoT architecture for low cost air pollution monitoring systems: a comprehensive system analysis, design considerations & development. *Sensors.* 18(9), 3021 (2018). <https://doi.org/10.3390/s18093021>
23. Duangsuwan, S., Takarn, A., Jamjareegularn, P.: A development on air pollution detection sensors based on NB-IoT network for smart cities. In: 2018 18th International Symposium on Communications and Information Technologies (ISCIT), pp. 313–317. IEEE (2018)
24. Aamer, H., et al.: A very low cost, open, wireless, internet of things (IoT) air quality monitoring platform. In: 2018 15th International Conference on Smart Cities: Improving Quality of Life Using ICT & IoT (HONET-ICT), pp. 102–106. IEEE (2018)
25. Mishra, L., Varma, S., others: Measurement of air quality index using internet of things. In: 2019 International Conference on Electrical, Electronics and Computer Engineering (UPCON), vol. 1–5. IEEE (2019)
26. Kumar, A., Kumari, M., Gupta, H.: Design and analysis of IoT based air quality monitoring system. In: 2020 International Conference on Power Electronics & IoT Applications in Renewable Energy and its Control (PARC), pp. 242–245. IEEE (2020)
27. Zhou, M., et al.: Internet of things (IoT) enabled smart indoor air quality monitoring system. In: Proceedings of the 2020 International Conference on Computing, pp. 89–93. Networks and Internet of Things (2020)
28. Shah, J., Mishra, B.: IoT-enabled low power environment monitoring system for prediction of PM_{2.5}. *Pervasive Mob. Comput.* 67, 101175 (2020). <https://doi.org/10.1016/j.pmcc.2020.101175>
29. Ayyasamy, A., et al.: AVRIM: adaptive void recovery mechanism to reduce void nodes in wireless sensor networks. *Peer-to-Peer Networking and Applications.* 13(3), 987–1001 (2020). 1–15. <https://doi.org/10.1007/s12083-019-00865-6>
30. Kalia, P., Ansari, M.A.: IOT based air quality and particulate matter concentration monitoring system. *Mater. Today Proc.* 32, 468–475 (2020). <https://doi.org/10.1016/j.matpr.2020.02.179>
31. Xu, Y., Liu, H., Duan, Z.: A novel hybrid model for multi-step daily AQI forecasting driven by air pollution big data. *Air Quality, Atmosphere & Health.* 13(2), 197–207 (2020). <https://doi.org/10.1007/s11869-020-00795-w>
32. Li, Y., et al.: Air Quality Forecasting with Artificial Intelligence Techniques: A Scientometric and Content Analysis. *Environmental Modelling & Software.* 105329 (2022)
33. Han, J., et al.: Semi-supervised air quality forecasting via self-supervised hierarchical graph neural network. *IEEE Trans. Knowl. Data Eng.*, 1 (2022). <https://doi.org/10.1109/tkde.2022.3149815>
34. Siddique, A.B., et al.: An intelligent and secure air quality monitoring system using neural network algorithm and blockchain. *IETE J. Res.*, 1–14 (2022). <https://doi.org/10.1080/03772063.2022.2052984>
35. Van, N., et al.: A new model of air quality prediction using lightweight machine learning. *Int. J. Environ. Sci. Technol.*, 1–12 (2022). <https://doi.org/10.1007/s13762-022-04185-w>
36. Benifa, J., Kumar, P.D., Rose, J.: Prediction of air quality index using machine learning techniques and the study of its influence on the health hazards at urban environment. In: *Computational Intelligence Techniques For Green Smart Cities* Springer, pp. 249–269 (2022)
37. Walczak, S.: Artificial neural networks. In: *Advanced Methodologies and Technologies in Artificial Intelligence, Computer Simulation, and Human-Computer Interaction* IGI Global, pp. 40–53 (2019)
38. McCulloch, W.S., Pitts, W.: A logical calculus of the ideas immanent in nervous activity. *Bull. Math. Biophys.* 5(4), 115–133 (1943). <https://doi.org/10.1007/bf02478259>
39. Hebb, D.O.: *The Organization of Behavior: A Neuropsychological Theory.* Science editions (1949)
40. Quek, S.G., et al.: Multi-attribute multi-perception decision-making based on generalized T-spherical fuzzy weighted aggregation operators on neutrosophic sets. *Mathematics.* 7(9), 780 (2019). <https://doi.org/10.3390/math7090780>
41. Gholami, R., Fakhari, N.: Support vector machine: principles, parameters, and applications. In: *Handbook of Neural Computation* Elsevier, pp. 515–535 (2017)
42. Martínez-Ramón, M., Christodoulou, C.: Support vector machines for antenna array processing and electromagnetics. *Synthesis Lectures on Computational Electromagnetics.* 1(1), 1–120 (2005). <https://doi.org/10.2200/s00020ed1v01y200604cem005>
43. Wang, L.: *Support Vector Machines: Theory and Applications*, vol. 177. Springer Science & Business Media (2005)
44. Tuan, T.A., et al.: Performance evaluation of Botnet DDoS attack detection using machine learning. *Evolutionary Intelligence.* 13(2), 1–12 (2019). <https://doi.org/10.1007/s12065-019-00310-w>
45. Enriko, I.K.A., Suryanegara, M., Gunawan, D.: Heart disease prediction system using k-nearest neighbor algorithm with simplified patient's health parameters. *J. Telecommun. Electron. Comput. Eng.* 8(12), 59–65 (2016)
46. García, V., Debreuve, E., Barlaud, M.: Fast k nearest neighbor search using GPU. In: 2008 IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops, pp. 1–6. IEEE (2008)
47. Patro, S.G.K., et al.: A hybrid action-related K-nearest neighbour (HAR-KNN) approach for recommendation systems. *IEEE Access.* 8, 90978–90991 (2020). <https://doi.org/10.1109/access.2020.2994056>
48. Miner, L., et al.: *Practical Predictive Analytics and Decisioning Systems for Medicine: Informatics Accuracy and Cost-Effectiveness for Healthcare Administration and Delivery Including Medical Research.* Academic Press (2014)
49. Shi, X., et al.: An accident prediction approach based on XGBoost. In: 2017 12th International Conference on Intelligent Systems and Knowledge Engineering (ISKE), pp. 1–7. IEEE (2017)

50. Chen, T., Guestrin, C.: Xgboost: a scalable tree boosting system. In: *Proceedings of the 22nd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining*, pp. 785–794 (2016)
51. Chellapilla, K., Puri, S., Simard, P.: High performance convolutional neural networks for document processing. In: *Tenth International Workshop on Frontiers in Handwriting Recognition* (2006)
52. Hochreiter, S., Schmidhuber, J.: Long short-term memory. *Neural Comput.* 9(8), 1735–1780 (1997). <https://doi.org/10.1162/neco.1997.9.8.1735>
53. Jha, S., et al.: Deep learning approach for software maintainability metrics prediction. *IEEE Access.* 7, 61840–61855 (2019). <https://doi.org/10.1109/access.2019.2913349>
54. Kim, T.Y., Cho, S.B.: Predicting residential energy consumption using CNN-LSTM neural networks. *Energy.* 182, 72–81 (2019). <https://doi.org/10.1016/j.energy.2019.05.230>
55. Huang, C.J., Kuo, P.H.: A deep cnn-lstm model for particulate matter (PM_{2.5}) forecasting in smart cities. *Sensors.* 18(7), 2220 (2018). <https://doi.org/10.3390/s18072220>
56. Zhang, Y., et al.: YNU-HPCC at EmoInt-2017: using a CNN-LSTM model for sentiment intensity prediction. In: *Proceedings of the 8th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, pp. 200–204 (2017)
57. Azeez, N.A., et al.: Windows PE malware detection using ensemble learning. In: *Informatics.* 8, pp. 10. Multidisciplinary Digital Publishing Institute. (2021)
58. Pelikan, M., et al.: The Bayesian optimization algorithm. In: *Proceedings of the Genetic and Evolutionary Computation Conference GECCO-99*, 1, pp. 525–532. Citeseer. (1999)
59. Li, H., et al.: Novel analysis–forecast system based on multi-objective optimization for air quality index. *J. Clean. Prod.* 208, 1365–1383 (2019). <https://doi.org/10.1016/j.jclepro.2018.10.129>
60. Amini, H., et al.: Short-term associations between daily mortality and ambient particulate matter, nitrogen dioxide, and the air quality index in a Middle Eastern megacity. *Environ. Pollut.* 254, 113121 (2019). <https://doi.org/10.1016/j.envpol.2019.113121>
61. Zheng, M., et al.: Seasonal trends in PM_{2.5} source contributions in Beijing, China. *Atmos. Environ.* 39(22), 3967–3976 (2005). <https://doi.org/10.1016/j.atmosenv.2005.03.036>
62. Hua, C., et al.: Impact of dust weather radiation on air quality and solar. *Meteorological and Environmental Research* 10(4), 11–22 (2019)
63. Brook, R.D., et al.: Particulate matter air pollution and cardiovascular disease: an update to the scientific statement from the American Heart Association. *Circulation* 121(21), 2331–2378 (2010). <https://doi.org/10.1161/cir.0b013e3181d8ce1>
64. Aghenta, L.O., Iqbal, M.T.: Low-cost, open source IoT-based SCADA system design using thinger. IO and ESP32 thing. *Electronics.* 8(8), 822 (2019). <https://doi.org/10.3390/electronics8080822>
65. AirNow.gov. <https://www.airnow.gov/>. (Accessed on 11/02/2022)
66. You, I., et al.: Misbehavior detection of embedded IoT devices in medical cyber physical systems. In: *Proceedings of the 2018 IEEE/ACM International Conference on Connected Health: Applications, pp. 88–93. Systems and Engineering Technologies* (2018)
67. Sharma, V., et al.: Behavior and vulnerability assessment of drones-enabled industrial internet of things (IIoT). *IEEE Access.* 6, 43368–43383 (2018). <https://doi.org/10.1109/access.2018.2856368>
68. Sharma, V., You, I., Guizani, N.: Security of 5G-V2X: technologies, standardization, and research directions. *IEEE Network.* 34(5), 306–314 (2020). <https://doi.org/10.1109/mnet.001.1900662>
69. Sharma, V., et al.: Security, privacy and trust for smart mobile-Internet of Things (M-IoT): a survey. *IEEE Access.* 8, 167123–167163 (2020). <https://doi.org/10.1109/access.2020.3022661>
70. Sharma, V., et al.: BRIoT: behavior rule specification-based misbehavior detection for IoT-embedded cyber-physical systems. *IEEE Access.* 7, 118556–118580 (2019). <https://doi.org/10.1109/access.2019.2917135>
71. Khan, R., et al.: A machine learning based energy-efficient non-orthogonal multiple access scheme. In: *14th International Forum on Strategic Technology (IFOST-2019)*, pp. 330–335. —TomskTomsk, Russia (2019). [proceedings]
72. Sharma, V., et al.: MIH-SPFP: MIH-based secure cross-layer handover protocol for Fast Proxy Mobile IPv6-IoT networks. *J. Netw. Comput. Appl.* 125, 67–81 (2019). <https://doi.org/10.1016/j.jnca.2018.09.002>
73. Firebase. <https://firebase.google.com/>. (Accessed on 11/02/2022)
74. Crockford, D.: The Application/json Media Type for Javascript Object Notation (Json) (2006). RFC 4627
75. UCI Machine Learning Repository: Beijing PM_{2.5} Data Set. <https://archive.ics.uci.edu/ml/datasets/Beijing+PM2.5+Data>. (Accessed on 11/02/2022)
76. Liang, X., et al.: Assessing Beijing's PM_{2.5} pollution: severity, weather impact, APEC and winter heating. *Proc. Math. Phys. Eng. Sci.* 471(2182), 20150257 (2015). <https://doi.org/10.1098/rspa.2015.0257>
77. AQI India Dataset. <https://www.kaggle.com/>. Accessed on 03 02 2021
78. Tanwar, S., et al.: Human arthritis analysis in fog computing environment using Bayesian network classifier and thread protocol. *IEEE Consumer Electronics Magazine.* 9(1), 88–94 (2019). <https://doi.org/10.1109/mce.2019.2941456>
79. Python Library for the Statistical Parameters. <https://scikit-learn.org/stable/supervised%5Flearning.html%23supervised%2Dlearning>. (Accessed on 11/02/2022)

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